

## Education

**University of Michigan - Ann Arbor**  
B.S.E. in Computer Science Engineering

**Ann Arbor, MI**  
Aug. 2021 - May 2025

**Coursework:** DS&A, Computer Organization, Software Engineering, Web Systems, Operating Systems (6 credit), Machine Learning, Calculus (I, II, & IV), Physics (I & II), Statistics | Game Design & Development, Network Systems (Ongoing)

## Related Experience

**Orientation Leader | University of Michigan**

**Ann Arbor, MI**  
May. 2024 - September 2024

- Coordinated daily orientations for 1000-1500 attendees, demonstrating strong organizational and communication skills.
- Delivered and planned presentations to large groups, showcasing public speaking abilities and the aptitude to convey complex information clearly.
- Facilitated small group activities and Q&A sessions, utilizing digital tools to enhance engagement and technical proficiency.
- Ensured safety and smooth operations by managing logistics, verifying identities, and maintaining accurate records.

## Projects

**Network File System | C++**

- Implemented a multithreaded network file system, enabling efficient handling of simultaneous file operations through a manager-worker threading model.
- Enabled thread-safe file and directory operations employing mutexes and upgradeable locks to ensure data integrity while optimizing concurrent access.
- Designed and tested the system for compatibility with both initialized and new file systems, ensuring robust performance and versatility in different operational states.
- Rigorously tested for stability and concurrency control, resulting in the system's ability to maintain consistent performance and reliability, even with high volumes of client requests.

**Memory Management Pager | C++**

- Implemented a memory management pager for an OS simulation that handles process creation, utilizing copy-on-write semantics to efficiently manage the duplication of parent process memory for child processes.
- Developed mapping function to extend process memory arenas with both swap-backed and file-backed pages, incorporating eager swap reservation and file sharing to maintain data integrity and system performance.
- Designed and managed page fault handling for read/write access violations, and created a cleanup routine to release resources associated with terminated processes.

**Thread Library | C++**

- Developed a multi-threading library in C++, enabling efficient thread management and execution across multiple CPUs.
- Implemented core threading constructs including thread creation, mutex locks, and condition variables to simulate concurrent processing and ensure thread-safety.
- Engineered context switching mechanisms and CPU scheduling, significantly improving the simulation of multi-threaded environments while enhancing understanding of concurrency and demonstrating problem-solving skills.

**Search Engine | Python, HTML5, JavaScript, CSS, SQLite3**

- Engineered an inverted index of MapReduce programs, enabling output of key data metrics such as inverse document frequency, term frequency, and document normalization factor.
- Designed an index server to process user queries and deliver search results in JSON format, ranked by PageRank score; further enhanced the system by developing an additional search server that ingests data from the index server to provide Google-like page results, creating an optimized user experience.
- Effectively employed multithreading and distributed systems, demonstrating proficiency in handling complex computational tasks.

**Insta485 | Python, HTML5, JavaScript, CSS, SQLite3**

- Led the full-stack development of an Instagram-inspired application, utilizing Python's Flask library for comprehensive user operations and employing JavaScript and React to enhance dynamic features such as infinite scrolling and real-time rendering.
- Managed diverse data sets pertaining to user accounts, interaction, and posts using robust SQL database solutions, contributing to a seamless and interactive user experience.

## Skills

**Tools & Languages:** C, C++, C#, Python, Java, JavaScript, HTML, CSS, SQL, Unity, Git, GitHub